Retail Sales Forecasting Using SARIMA

Time Series Analysis & SARIMA Modeling

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Project Overview

Dataset

48 months of simulated monthly retail sales (Jan 2020 – Dec 2023). Fields: SalesAmount, Promotion (binary), HolidayMonth (binary). No missing values.

Objective

Identify trend and seasonality, evaluate stationarity and autocorrelation, fit SARIMA, and produce a 6-month forecast (Jan–Jun 2024).

Tools & Workflow

Python ecosystem: Pandas, NumPy, Statsmodels, Scikit-learn. Steps: EDA → Decomposition → Stationarity tests → SARIMA selection → Diagnostics → Forecasting.

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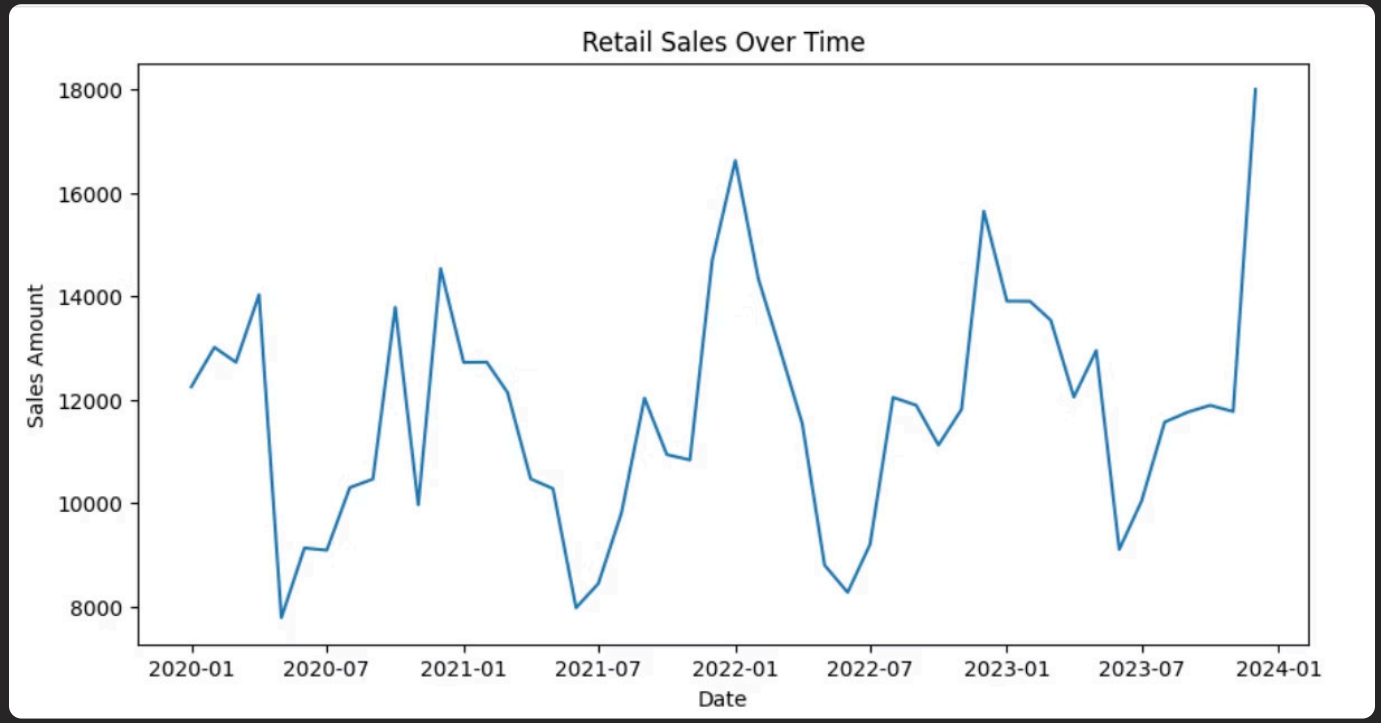
Data Exploration & Key Patterns



Range: ~8,000 to ~18,000 units/month.

Trend: Upward slope ≈ +50 units/month on average — persistent growth across 2020–2023.

Seasonality: Strong Q4 peaks (holiday effect); mid-year troughs recurring each year.

Promotions: Promotional months show +10–30% uplift; interact multiplicatively with holiday months (largest spikes).

Implication: Model must represent additive trend + strong annual seasonality and occasional exogenous spikes.

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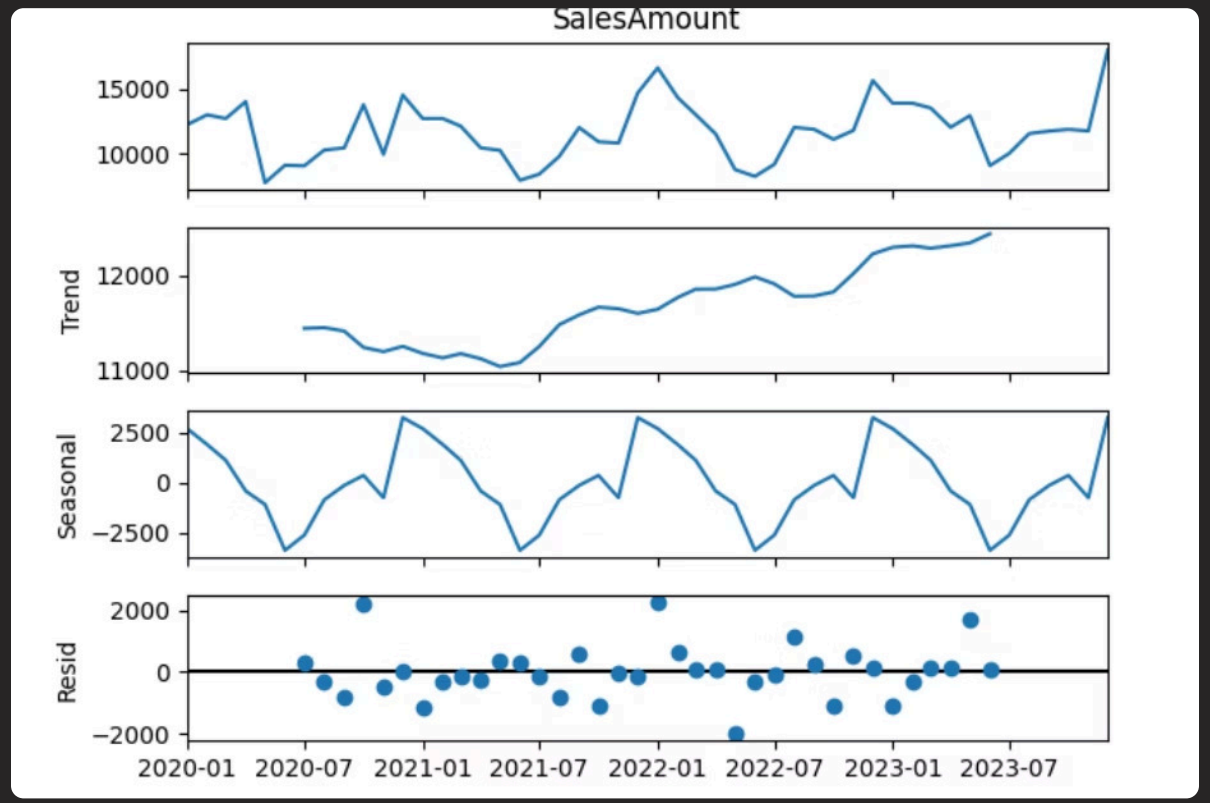
Seasonal Decomposition (Additive, period=12)

Decomposition highlights clear components:

- **Trend:** Gradual rise from ~10,500 (2020) to ~12,500 (2023).
- **Seasonal:** Monthly pattern oscillating roughly ±2,500 around the trend; consistent annual shape with Dec peak.
- **Residuals:** Centered near zero, largely random; a few pronounced positive outliers align with promotion+holiday months.



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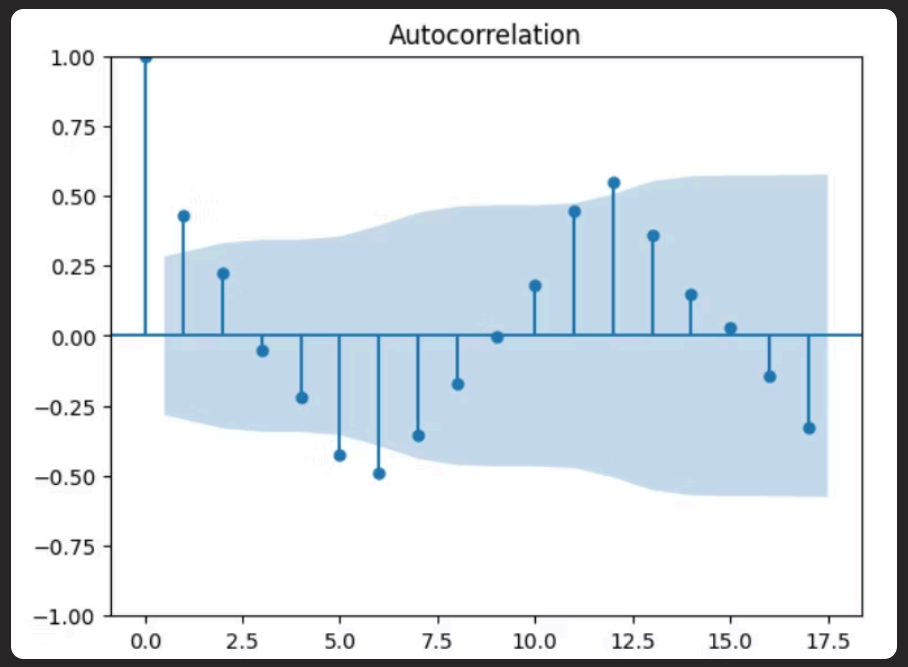
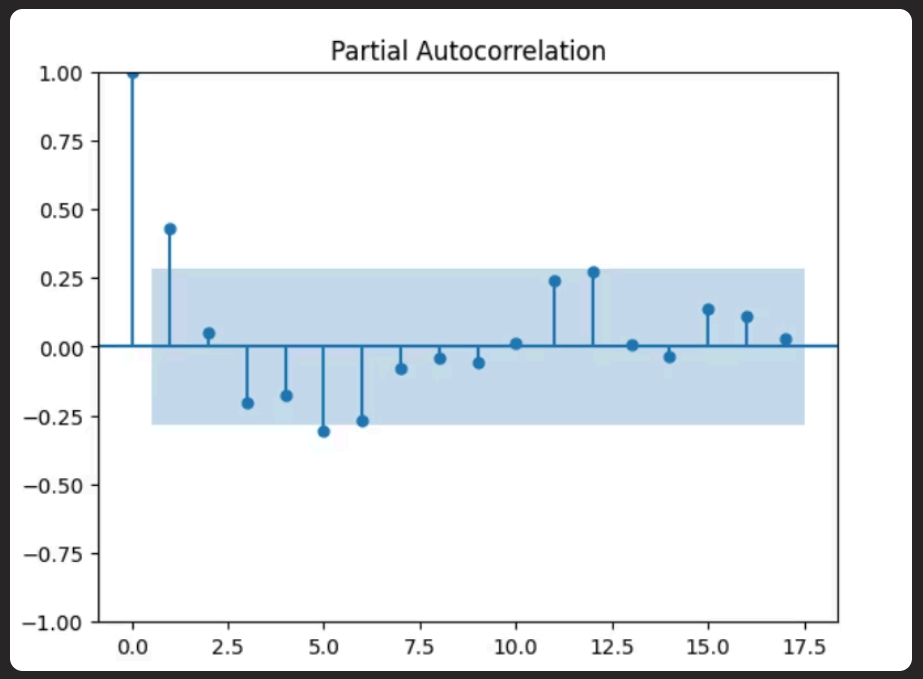
Stationarity & Autocorrelation



ADF Test: Test statistic = −4.514, p = 0.000185 → Series considered **stationary** (no nonseasonal differencing required, d = 0).

Interpretation: Deterministic trend removed or negligible after preprocessing; seasonal structure remains.



ACF: Significant at lags 1,2 and 12 → seasonal correlation at 12 months.

PACF: Sharp spike at lag 1 then cutoff → suggests AR(1) component; seasonal spikes at lag 12 suggest seasonal AR/MA terms.

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Model Selection — SARIMA Specification

Chosen Model

SARIMA(1,0,1)(1,1,1)[12]

Rationale

- p=1: PACF spike at lag 1
- d=0: ADF indicates stationarity
- q=1: ACF residual structure at lag 1
- P=1, D=1, Q=1, s=12: seasonal spikes at lag 12 and observed annual cycle; seasonal differencing removes yearly level shifts

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Model Diagnostics

Training / Test Split: 38 months train | 10 months test (≈80/20).

Significant Parameters: Seasonal AR (ar.S.L12) = −0.689, $p < 0.001$ — captures annual inertia.

Goodness & Residual Checks:

- Ljung-Box $p = 0.50$ → no evidence against residual whiteness (uncorrelated).
- Jarque-Bera $p = 0.52$ → residuals approximately normal.
- Heteroskedasticity $p = 0.45$ → variance stable.
- Information Criteria: AIC = 204.9, BIC = 207.3.

Diagnostics indicate a well-specified model: residuals resemble white noise and are approximately Gaussian — suitable for forecasting.

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Forecast Performance on Test Set

Evaluation Metrics (test set)

Key insight: Model handles baseline trend and seasonality well; promotional + holiday compound spikes (e.g., Dec 2023) are slightly underestimated without exogenous inputs.

- **MAE**

  1,319.75 — average absolute error ≈ 1.32k units/month

- **RMSE**

  1,814.58 — penalizes larger misses

- **MAPE**

  10.39% — ~10% average deviation, acceptable for monthly retail forecasts

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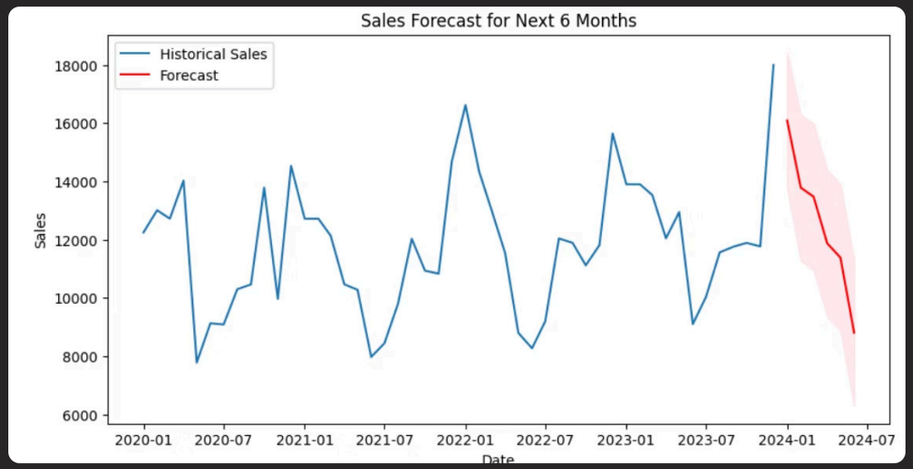
6-Month Forecast (Jan–Jun 2024)



Forecast table (95% CI)

Month	Forecast	95% CI
Jan 2024	16,093	13,755 — 18,430
Feb 2024	13,785	11,286 — 16,284
Mar 2024	13,478	10,964 — 15,991
Apr 2024	11,875	9,360 — 14,390
May 2024	11,385	8,870 — 13,900
Jun 2024	8,810	6,295 — 11,325

Interpretation: Forecast shows seasonal decline from Jan peak toward mid-year trough — consistent with historical patterns. Confidence bands widen slightly mid-year due to seasonal variability and occasional promotions.

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Conclusions & Recommendations

Key Findings

SARIMA(1,0,1)(1,1,1)[12] captures trend and annual seasonality; test MAPE = 10.39% is acceptable for operational planning.

Immediate Recommendations

- Adopt SARIMAX to include Promotion and Holiday flags as exogenous regressors to better model spike events.
- Extend historical data to ≥5 years for more robust seasonal estimation and parameter stability.
- Run automated grid search (auto_arima) and consider ensembling with Prophet for improved robustness.

Operational Next Steps

1. Integrate promotion calendar and planned marketing events as exogenous inputs.
2. Deploy rolling retrain schedule (monthly) and monitor forecast error drift.
3. Implement alerting for large residuals indicating atypical promotional impact.

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